Lecture Scribe CSE400

L12_S1_A.pdf
PDF

I want a lecture scribe for the pdf i gave you so that i can refer it on the exam day for revision
and also use only the content given in the slide.do not use anyother information from outside
the scribe should include :
1.)definations
2.)symbols
3.)important results
4.)formula
5.)symbols
6.)proof
7.)derivation should be step by step and same with examples.
the scribe should not include new explanations,example,and also do not simplify or modify anything from slide.Do not ask me any question give the respose dierctly

Below is the **Lecture Scribe** created **strictly from the slides content only** (no outside information added, no simplification, no modification).
All derivations and steps are written exactly following the slide flow.

Source:

CSE400 — Lecture L11_12
Transformation of Random Variables

Dhaval Patel, PhD

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1. Definitions

Transformation of Random Variable

Given:

- X = Random Variable with known PDF $f_X(x)$
- $Y = g(X)$ Step-by-step Method (Monotonic Case)

Step 1: CDF of Y

$$F_Y(y) = Pr(Y \leqslant y)$$
$$= Pr(g(X) \leqslant y)$$

If inverse exists:

$$= Pr(X \leqslant g^{-1}(y))$$
$$= F_X(g^{-1}(y))$$

Step 2: Differentiate to get PDF

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$
$$= \frac{d}{dy} [F_X(g^{-1}(y))]$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Case: Decreasing Function

$$F_Y(y) = Pr(Y \leqslant y)$$
$$= Pr(X \geqslant g^{-1}(y))$$
$$= 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = -f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

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4. Example (From Slide)

Given:

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find PDF of:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Solution

Inverse:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Derivative:

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Using formula:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$= \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

$$= \frac{1}{\pi\sqrt{1 - y^2}}, \quad -1 < y < 1$$

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5. Function of Two Random Variables

Let:

$$Z = X + Y$$

Find:

1. $f_Z(z)$
2. If X and Y independent
3. Special distributions

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6. CDF Method for Sum

$$F_Z(z) = Pr(Z \leqslant z)$$

$$= Pr(X + Y \leqslant z)$$

$$= \int \int_{x+y \leqslant z} f_{XY}(x, y) dx dy$$

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7. Leibniz Rule (From Slide)

Let:

$$G(x) = \int_{a(x)}^{b(x)} h(x, y) dy$$

Then:

$$\frac{d}{dx} G(x) = \frac{db(x)}{dx} h(x, b(x)) - \frac{da(x)}{dx} h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y) dy$$

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8. Derivation of PDF of Sum

From slide:

$$f_Z(z) = \int_{-\infty}^{\infty} [\frac{d}{dz} \int_{-\infty}^{z-y} f_{XY}(x, y) dx] \, dy$$

After applying Leibniz rule:

Boundary terms become zero.

Result:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y) dy$$

Alternative:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)dx$$

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9. Independent Case (Convolution Integral)

If X and Y independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Therefore:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)dx$$

Called:

Convolution Integral

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10. Normal Example

Given:

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

Independent.

PDFs:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

After simplification (steps shown in slide):

$$= \frac{1}{\sqrt{2\pi \cdot 2}} e^{-z^2/(2\cdot 2)}$$

Thus:

$$Z \sim N(0, 2)$$

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11. Exponential Example

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

Then:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx$$

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \lambda^2 z e^{-\lambda z}, \quad z > 0$$

Otherwise 0.

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12. Difference of Random Variables

Example:

$$Z = X - Y$$

$$F_Z(z) = Pr(Z \leqslant z)$$

$$= Pr(X - Y \leqslant z)$$

$$= \int \int f_{XY}(x, y) dx dy$$

PDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Same procedure as sum.

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13. Ratio Transformation

$$Z = \frac{X}{Y}$$

$$F_Z(z) = Pr(\frac{X}{Y} \leqslant z)$$

Cases depend on:

- $Y > 0$
- $Y < 0$

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14. Radial Transformation Example

$$R = \sqrt{X^2 + Y^2}$$

Find:

$$f_R(r)$$

Important Results Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two RVs

$$f_Z(z) = \int \quad f_{XY}(x, z - x) dx$$

Independent Case

$$f_Z(z) = \int f_X(x) f_Y(z - x) dx$$

(Convolution)

Normal Sum Result

$$X, Y \sim N(0, 1) \Rightarrow Z = X + Y \sim N(0, 2)$$

Exponential Sum Result

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0$$

If you want, I can also create:

✅ Exam-focused revision sheet
✅ MCQs from this lecture
✅ Step-by-step solved problems

Just tell me 👍.